

Point

From Number Theory to Algebraic Geometry and Logic

Session S4 — Category Theory Essentials

Complete lecture notes with proofs, examples, and exercises

Abstract

This session installs the *grammar* in which the rest of the course is written: categories, functors, natural transformations, limits and colimits (computed concretely: inverse limits give $\mathbb{Z}_p$, profinite groups, and $G_{\mathbb{Q}}$; directed colimits give stalks), adjunctions (free $\dashv$ forgetful, tensor $\dashv$ Hom, $f^{-1} \dashv f_*$), and the Yoneda lemma — the precise form of the slogan “an object is determined by its relationships”, which becomes in S29 “a scheme is determined by its points over all bases” and in S54 “a point of a topos is a morphism into it”. Proofs are complete; the geometric payoff of Yoneda is deferred to S29 by design.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: universal properties as the course’s grammar; where S4 feeds
0:10–0:40	§1 Categories, functors, natural transformations
0:40–1:10	§2 Limits and colimits, concretely: inverse limits ($\mathbb{Z}_p$, $G_{\mathbb{Q}}$), directed colimits (stalks)
1:10–1:20	<i>Break</i>
1:20–1:45	§3 Adjunctions; adjoints preserve (co)limits; right-exactness as a shadow
1:45–2:10	§4 Yoneda lemma (proof); embedding; optional remark on monads (S27)
2:10–2:30	Exercises / wrap-up: object = its network of maps

Conventions

Categories are denoted $\mathcal{A}, \mathcal{B}, \mathcal{C}$; $\mathcal{C}^{\text{op}}$ is the opposite category; $[\mathcal{A}, \mathcal{B}]$ the functor category. We work with small-category shortcuts freely (size issues are irrelevant for this course). All (co)limits are defined by universal properties; concrete constructions are given in **Set** and **$R\text{-Mod}$** .

Contents

1	Categories, Functors, Natural Transformations	2
2	Limits and Colimits, Concretely	2
3	Adjunctions	4
4	The Yoneda Lemma (and a Remark on Monads)	4
5	Exercises	5
	Where This Session Feeds the Course	5

1 Categories, Functors, Natural Transformations

Definition 1.1 (Category). A **category** $\mathcal{C}$ is a class of **objects** together with, for each pair (A, B) , a set $\text{Hom}_{\mathcal{C}}(A, B)$ of **morphisms**, an associative composition $\text{Hom}(B, C) \times \text{Hom}(A, B) \rightarrow \text{Hom}(A, C)$, and identities $1_A \in \text{Hom}(A, A)$ acting as units.

Example 1.2 (Categories we live in). • **Set**, **Grp**, **Ring**, **Top**, and **$R\text{-Mod}$** (S2): morphisms = maps, homomorphisms, continuous maps, R -linear maps.

- A poset $(P, \leq)$ as a category: one morphism $x \rightarrow y$ iff $x \leq y$ (composition = transitivity); e.g. $(\mathbb{N}, |)$ and the poset of open sets $\text{Open}(X)$ (S26).
- A monoid M as a one-object category; a group as a one-object category with all morphisms invertible (P1).
- The opposite category $\mathcal{C}^{\text{op}}$: same objects, arrows reversed.

Definition 1.3 (Functor). A **covariant functor** $F : \mathcal{A} \rightarrow \mathcal{B}$ assigns objects to objects and morphisms to morphisms with $F(g \circ f) = F(g) \circ F(f)$ and $F(1_A) = 1_{F(A)}$; a **contravariant functor** $\mathcal{A} \rightarrow \mathcal{B}$ is a covariant functor $\mathcal{A}^{\text{op}} \rightarrow \mathcal{B}$.

Example 1.4 (Functors of the course). • Forgetful functors $U : \mathbf{Ring} \rightarrow \mathbf{Set}$, $U : R\text{-Mod} \rightarrow \mathbf{Set}$, $U : \mathbf{Top} \rightarrow \mathbf{Set}$.

- Free functor $F : \mathbf{Set} \rightarrow R\text{-Mod}$, $X \mapsto R^{(X)}$ (S2).
- Representable functors $\text{Hom}(A, -) : \mathcal{A} \rightarrow \mathbf{Set}$ and $\text{Hom}(-, B) : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Set}$.
- Fundamental group $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$ (P3).
- **Spectrum** $\text{Spec} : \mathbf{Ring}^{\text{op}} \rightarrow \mathbf{Top}$: a ring map $\varphi : R \rightarrow S$ sends a prime $\mathfrak{q} \subseteq S$ to the prime $\varphi^{-1}(\mathfrak{q}) \subseteq R$ (S1); continuity checked in S28. Contravariance is the first appearance of “points pull back”.

Definition 1.5 (Natural transformation). Given functors $F, G : \mathcal{A} \rightarrow \mathcal{B}$, a **natural transformation** $\eta : F \Rightarrow G$ is a family $\eta_A : F(A) \rightarrow G(A)$ such that for every $f : A \rightarrow B$ the square $\eta_B \circ F(f) = G(f) \circ \eta_A$ commutes (**naturality**).

Example 1.6. • Abelianization: $\text{ab}_G : G \rightarrow G^{\text{ab}} = G/[G, G]$ is natural in G (P1).

- Double dual: $D_V : V \rightarrow V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is natural on finite-dimensional k -vector spaces (and D_V is then an isomorphism); for infinite-dimensional V it is natural but not an isomorphism.
- Identities and (vertical/horizontal) compositions of natural transformations make $[\mathcal{A}, \mathcal{B}]$ a category, the **functor category**.

Remark 1.7 (Isomorphism vs equivalence). An **equivalence** of categories ($F \dashv G$ with unit and counit natural *isomorphisms*) is the correct notion of “same”; it appears in S27 ($\text{Sh}(X) \simeq \text{étale spaces over } X$) and S53 (topoi). Isomorphism of categories is too rigid and rarely holds in practice.

2 Limits and Colimits, Concretely

Definition 2.1 (Cone, limit, colimit). A **cone** over a diagram $D : I \rightarrow \mathcal{C}$ with vertex L is a family $L \rightarrow D(i)$ commuting with all diagram arrows; the **limit** $\lim D$ is a universal cone (every cone factors through it uniquely). **Colimits** $\text{colim } D$ are cones under D (arrows $D(i) \rightarrow L$), universal dually.

Example 2.2 (The zoo). • Limits: terminal object; products $\prod_i A_i$; equalizers $\{a \in A : f(a) = g(a)\}$; **pullbacks** (fiber products) of $A \rightarrow C \leftarrow B$ — the categorical form of the fiber products of schemes in S30 and of the sheaf-equalizer axiom in S26.

- Colimits: initial object; coproducts $\bigsqcup_i A_i$; coequalizers; pushouts.

Proposition 2.3 (Constructions in **Set** and **R-Mod**). *Limits exist in **Set** and **R-Mod** and are computed as subsets of products ($\lim D = \{(x_i) \in \prod_i D(i) : \text{compatible}\}$); colimits exist and are quotients of coproducts by the identification relations imposed by the diagram arrows. In particular limits and colimits are unique up to unique isomorphism (universal property).*

Proof. Direct verification of the universal properties: a cone over D with vertex L is exactly a family $(L \rightarrow D(i))_i$ landing in the compatible subset, uniquely; dually a cocone is a family from the coproduct killing the relations, uniquely. Uniqueness of (co)limits: two universal objects admit maps both ways whose composites must be identities by uniqueness in the universal property. $\square$

Definition 2.4 (Inverse (projective) limit). An **inverse system** is a functor from a directed poset $(I, \leq)$ viewed with arrows $i \rightarrow j$ for $i \leq j$ reversed; its limit $\varprojlim_i A_i$ is the set of compatible families $(a_i)_i$ with $\varphi_{ij}(a_j) = a_i$ for $i \leq j$ (a special case of Proposition 2.3).

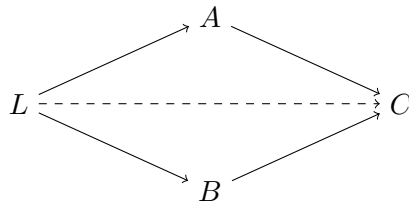
Example 2.5 (Inverse limits the course runs on). • $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ with the reduction maps (S18): an element is a compatible system of residues mod p^n for all n — “a point known to infinite p -adic precision”.

- Profinite groups $\varprojlim G_i$ over finite quotients (S34, P3 deck groups); in particular $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z}$.
- $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) = \varprojlim_{L/\mathbb{Q}} \text{Gal}(L/\mathbb{Q})$ over finite Galois subextensions (S16): an element of $G_{\mathbb{Q}}$ is a compatible system of finite symmetries.

Definition 2.6 (Directed (inductive) colimit). For a directed poset $(I, \leq)$ and system (A_i, φ_{ij}) , the **directed colimit** $\varinjlim_i A_i$ is the quotient of $\bigsqcup_i A_i$ by $a_i \sim \varphi_{ij}(a_i)$; elements are “eventual” equivalence classes.

Example 2.7 (Directed colimits the course runs on). • **Stalks:** for a presheaf $\mathcal{F}$ on X (S26), $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$: a germ is a section over some neighbourhood of x , identified when they agree on a smaller one — “the value of $\mathcal{F}$ at the point x ”.

- Localizations as colimits: $R[x^{-1}] = \varinjlim (R \xrightarrow{x} R \xrightarrow{x} \dots)$.
- Unions of directed families of submodules/subfields (S3: $\bar{\mathbb{Q}} = \bigcup$ finite extensions).



a cone over $A \rightarrow C \leftarrow B$; the universal one is the pullback (limit)

Figure 1: Limits are universal cones: compatible families, uniquely factoring. Colimits dualize: gluing by identification.

Remark 2.8 (Threads). A sheaf (S26) is precisely a presheaf turning certain colimits of opens into limits of sets (the equalizer axiom); étale spaces (S27) and topoi (S53) internalize this. Meanwhile “object determined by its (co)cones of maps” is the seed of Yoneda (§4) and hence of “point = morphism” (S29, S54).

3 Adjunctions

Definition 3.1 (Adjunction). An **adjunction** $F \dashv G$ between $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{A}$ is a family of bijections $\Phi_{A,B} : \text{Hom}_{\mathcal{B}}(F(A), B) \cong \text{Hom}_{\mathcal{A}}(A, G(B))$ natural in A and B . Equivalently: unit $\eta : 1_{\mathcal{A}} \Rightarrow GF$ and counit $\varepsilon : FG \Rightarrow 1_{\mathcal{B}}$ satisfying the triangle identities.

Example 3.2 (Adjunctions of the course). • Free $\dashv$ forgetful: $\text{Hom}_{R\text{-Mod}}(R^{(X)}, M) \cong \text{Hom}_{\text{Set}}(X, U(M))$ (S2).

- Abelianization $\dashv$ inclusion: $\text{Hom}_{\mathbf{Ab}}(G^{\text{ab}}, A) \cong \text{Hom}_{\mathbf{Grp}}(G, A)$.
- Tensor $\dashv$ Hom: $\text{Hom}_R(M \otimes_R N, P) \cong \text{Hom}_R(M, \text{Hom}_R(N, P))$ — the conceptual reason tensor is right-exact (Corollary 3.4); full use in S30.
- Sheaves: $f^{-1} \dashv f_*$ (S27); sheafification $a \dashv i$ (S27).

Proposition 3.3 (Adjoint preserve (co)limits). *A left adjoint preserves all colimits; a right adjoint preserves all limits.*

Proof. Let $F \dashv G$ and $C = \text{colim}_i D_i$ with cocone $D_i \rightarrow C$. For any $B \in \mathcal{B}$: $\text{Hom}_{\mathcal{B}}(F(C), B) \cong \text{Hom}_{\mathcal{A}}(C, G(B)) \cong \lim_i \text{Hom}_{\mathcal{A}}(D_i, G(B)) \cong \lim_i \text{Hom}_{\mathcal{B}}(F(D_i), B)$, naturally in B ; hence $F(C)$ represents the cocone functor of $(F(D_i))_i$, i.e. $F(C) \cong \text{colim}_i F(D_i)$ by uniqueness of colimits (Proposition 2.3). The limit case is dual. $\square$

Corollary 3.4 (Right-exactness as a shadow). *Forgetful functors preserve limits (products and equalizers of rings/modules are computed on underlying sets — consistent with S1–S2); free functors and $- \otimes_R N$ preserve colimits (coproducts and coequalizers) — which is exactly the right-exactness of tensor proved in S2, now explained: left adjoints preserve colimits but need not preserve limits, hence tensor can destroy kernels (the failure measured by Tor in S43).*

Definition 3.5 (Representable functor; universal element). $F : \mathcal{A} \rightarrow \mathbf{Set}$ is **representable** if $F \cong \text{Hom}_{\mathcal{A}}(A, -)$ for some A ; the image of 1_A is the **universal element**. Products, kernels, and limits are representable “cone functors”; this viewpoint is the gateway to Yoneda.

4 The Yoneda Lemma (and a Remark on Monads)

Theorem 4.1 (Yoneda lemma). *For any functor $F : \mathcal{A} \rightarrow \mathbf{Set}$ and object $A \in \mathcal{A}$, the map $\Psi : \text{Nat}(\text{Hom}(A, -), F) \rightarrow F(A)$, $\eta \mapsto \eta_A(1_A)$, is a bijection.*

Proof. Inverse map: given $a \in F(A)$, define $\eta_X^a : \text{Hom}(A, X) \rightarrow F(X)$ by $\eta_X^a(f) = F(f)(a)$. Naturality: for $g : X \rightarrow Y$, $\eta_Y^a(g \circ f) = F(g \circ f)(a) = F(g)(F(f)(a)) = F(g)(\eta_X^a(f))$. Then $\Psi(\eta^a) = \eta_A^a(1_A) = F(1_A)(a) = a$. Conversely, for any η and $f : A \rightarrow X$, naturality of η gives $\eta_X(f) = \eta_X(\text{Hom}(A, f)(1_A)) = F(f)(\eta_A(1_A)) = \eta_X^{\Psi(\eta)}(f)$, so $\eta = \eta^{\Psi(\eta)}$. Hence Ψ is bijective. $\square$

Corollary 4.2 (Yoneda embedding). *The functor $\mathcal{A} \rightarrow [\mathcal{A}^{\text{op}}, \mathbf{Set}]$, $A \mapsto \text{Hom}(-, A)$, is fully faithful; hence $A \cong B$ iff $\text{Hom}(-, A) \cong \text{Hom}(-, B)$. An object is determined by its network of maps into it — equivalently, by its “points” $X \rightarrow A$ over all bases X .*

Proof. Apply Theorem 4.1 with $F = \text{Hom}(-, B)$ on $\mathcal{A}^{\text{op}}$: $\text{Nat}(\text{Hom}(-, A), \text{Hom}(-, B)) \cong \text{Hom}(A, B)$, and composition respects this identification; fully faithfulness follows, and an isomorphism of representables yields an isomorphism of objects by standard diagram chasing (the images of $1_A, 1_B$ give mutually inverse maps). $\square$

Remark 4.3 (Where Yoneda becomes geometry). S29 makes Corollary 4.2 geometric: a scheme X is determined by its functor of points $T \mapsto X(T) = \text{Hom}(T, X)$; “ T -valued points” are morphisms, and non-representable functors are moduli problems (S41). S54 repeats the pattern logically: a point of a topos $\mathcal{E}$ is a morphism $\mathbf{Set} \rightarrow \mathcal{E}$. The proof above is all the machinery these deep theorems need.

Remark 4.4 (Optional: monads and comonads, in one paragraph). A **monad** on $\mathcal{C}$ is (T, η, μ) with $T : \mathcal{C} \rightarrow \mathcal{C}$ and natural $\eta : 1 \Rightarrow T$, $\mu : T^2 \Rightarrow T$ (associative, unital); e.g. the free-group monad on **Set**, and — the one that matters here — the sheafification monad $a \circ i$ on $\mathbf{PSh}(X)$, which is *idempotent* (μ an isomorphism): algebras over it are exactly sheaves (S27). Comonads dualize (e.g. forgetting-and-refreeing). Not used elsewhere in the course; included so the remark in S27 has a home.

5 Exercises

Exercise 1. Verify that a poset and a monoid, as in Example 1.2, satisfy the category axioms; describe functors between poset categories (monotone maps) and between one-object categories (monoid homomorphisms).

Exercise 2. Show $\mathrm{Hom}(A, -)$ is a covariant functor and $\mathrm{Hom}(-, B)$ a contravariant one; write out both actions on morphisms and check functoriality.

Exercise 3. Prove naturality of the double-dual map on finite-dimensional vector spaces and of abelianization (Example 1.6); give an example of a family of isomorphisms that is *not* natural.

Exercise 4. Compute products and equalizers in **Set** and in $R\text{-}\mathbf{Mod}$ explicitly and verify the universal properties; deduce Proposition 2.3 for these two shapes of diagram.

Exercise 5. Show $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n$ satisfies the universal property of the limit in **Ring** (compatible families of ring maps), and identify its elements with the p -adic series of S18.

Exercise 6. Describe the stalk $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$ concretely as germs (pairs (U, s) modulo agreement on smaller opens) and verify the universal property of the directed colimit.

Exercise 7. Prove free $\dashv$ forgetful for $R\text{-}\mathbf{Mod}$ by writing the unit and counit and checking the triangle identities; identify which adjunctions of Example 3.2 you can prove today.

Exercise 8. Prove directly (without Proposition 3.3) that $- \otimes_R N$ preserves coproducts and coequalizers; compare with the S2 right-exactness proof.

Exercise 9. Prove the Yoneda lemma (Theorem 4.1) again from scratch, then deduce that any two representations of a representable functor are isomorphic via a unique isomorphism compatible with the universal elements.

Exercise 10. Show $G \mapsto G^{\mathrm{ab}}$ is left adjoint to the inclusion $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$; write the unit and counit; explain why this makes abelianization a colimit (a coequalizer).

Exercise 11. Identify each of the following as a limit or a colimit, and of which shape: kernel of a ring map; cokernel; fiber product of schemes (S30); pushout of rings (gluing affines, S28); stalk; $\mathbb{Z}_p$; $\mathbb{Q}$ as a union of finite extensions.

Exercise 12 (Optional). Write down the free-monoid monad on **Set** explicitly (unit, multiplication) and verify the monad laws; then explain why its algebras are exactly monoids — the prototype of “algebras over a monad” and of the sheafification remark.

Where This Session Feeds the Course

- **Depends on:** S1 (primes for Spec), S2 (modules, tensor, exact sequences), P1 (groups, abelianization), P3 (π_1 as a functor).

- **Feeds:** S16, S34 (inverse limits: $G_{\mathbb{Q}}$, profinite groups); S26–S27 (presheaves as functors, stalks as colimits, sheafification adjunction and monad, étale-space equivalence); S28–S30 (Spec functoriality, fiber products as pullbacks); S29 (Yoneda, functor of points); S41 (moduli as non-representable functors); S53–S54 (topoi, geometric morphisms, points as morphisms); S43 (tensor’s left-adjointness as the reason for Tor).
- **Unifying thread:** universal properties say an object *is* its network of maps; Yoneda makes this a theorem; the course’s “point = morphism into the object” (S29, S54) is Yoneda applied to geometry and logic.

References for this session

- [1] Mac Lane, S. *Categories for the Working Mathematician*, ch.I–V (selected).
- [2] Leinster, T. *Basic Category Theory*, ch.1–6 (gentle, complete proofs).
- [3] Riehl, E. *Category Theory in Context* (free; excellent on Yoneda and adjunctions).

End of Session S4 — Act 0 complete; next: Act I, Session S5, Quadratic Residues and the Legendre Symbol